

USEPA DMR-QA 32

NPDES PERMITTEE DATA REPORT FORM



ERA

A Waters Company Quality

AUG 17 2012

Water Bureau

USEPA NPDES MI0020257

Permit #:

Permit Ext:

Permittee Name: Cadillac WWTP

Facility Address: 1121 Plett Rd.

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

Fax Number: (231) 775-2876

E-mail address: lab@cadillac-mi.net

For DMR-QA Study 32, conducted in 2012, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Yes ☒ No ☐

Submitted Complete and Accurate Data
by July 6, 2012

Yes ☒ No ☐

Received a Graded Report by July
27, 2012

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative
(as per 40 C.F.R. Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) Permit. Neither I nor any of my subordinates compared our results from independent analyses conducted by us or any other laboratory before we reported our results to the USEPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of
Certifying Official: Amy Vail

Title: Lab Manager

Signature:

Date Signed:

7/30/2012

Mailing Address: 200 N Lake Street

(enter only if different
from address above)

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

E-mail address: lab@cadillac-mi.net

United States

ENVIRONMENTAL PROTECTION AGENCYLaboratory Performance Evaluation
Laboratory DMR-QA Evaluation Study 32

USEPA NPDES Permit #: MI0020257

Permit Ext:

Mich. Dept. of Environmental Quality



AUG 17 2012

Water Bureau

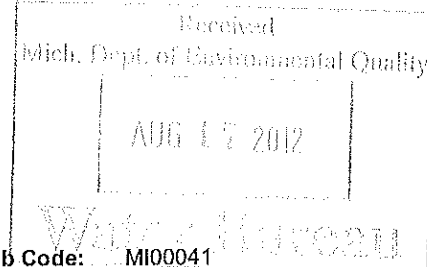
Identification of all CHEM, MICRO and TOX laboratories who did analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type
			CHEM	MICRO	TOX	
Cadillac WWTP	200 N. Lake St. Cadillac, MI 49601	MI00082	☒	☒	☐	G
Kar Labs	4425 Manchester Rd. Kalamazoo, MI 49001	MI00041	☒	☐	☐	C
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C

Lab Types: C = Commerical F = Federal G = Local Government I = Industrial O = Other S = State



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Permittee Name: Cadillac WWTP

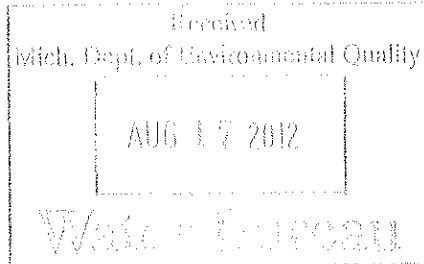
Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 32

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Minerals Conductivity at 25°C Chloride Potassium Sodium Fluoride Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C Alkalinity as CaCO ₃			
Hardness Non-Filterable Residue (TSS) Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃			
pH pH			
Settleable Solids Settleable Solids			
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Simple Nutrients Ammonia as N ortho-Phosphate as P Nitrate as N Nitrate + Nitrite as N			
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P			
Nitrite Nitrite as N			
Demand COD			



Permittee Name: Cadillac WWTP

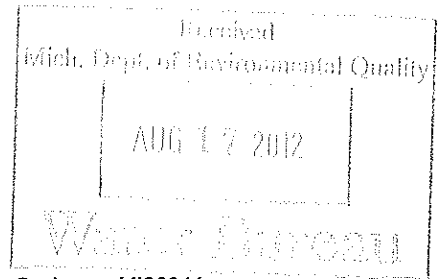
Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 32

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
CBOD TOC BOD			
<u>Oil & Grease Concentrate</u> n-Hexane Extractable Material(O&G) (Grav)			
<u>Oil & Grease</u> n-Hexane Extractable Material(O&G) (IR) n-Hexane Extractable Material(O&G) (Grav)			
<u>Trace Metals</u> Chromium Antimony Lead Manganese Molybdenum Aluminum Copper Beryllium Zinc Barium Cadmium Iron Silver Strontium Vanadium Boron Selenium Thallium Arsenic Cobalt Nickel			
<u>Mercury</u> Mercury			
<u>Low-Level Mercury</u> Low Level Mercury EPA 1631E	☐	☐	☒
<u>Hexavalent Chromium</u> Hexavalent Chromium			
<u>Turbidity</u> Turbidity			



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Chemistry/Microbiology Analyte Checklist

DMRQA Study 32

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
<u>Total Cyanide</u> Cyanide, total			
<u>Total Phenolics (4-AAP)</u> Phenolics, total			
<u>Total Residual Chlorine</u> Total Residual Chlorine			
<u>Low-Level Total Residual Chlorine</u> Low Level Total Residual Chlorine			
<u>WasteWatR™ Coliform MicrobE™</u> Total Coliforms (MF) Fecal Coliforms (MF) E.coli (MF) Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)			

Print Name

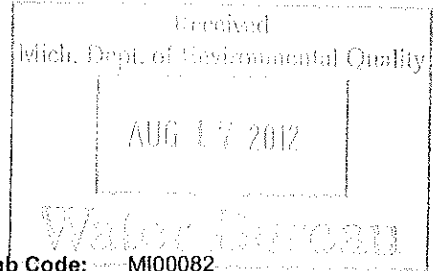
Signature/Title

Date

Use a separate checklist for EACH lab used



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Permittee Name: Cadillac WWTP

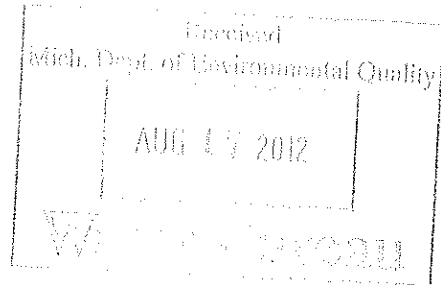
Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

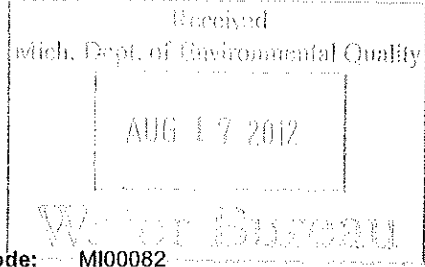
DMRQA Study 32

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Minerals Conductivity at 25°C Chloride Potassium Sodium Fluoride Total Dissolved Solids at 180°C Sulfate Total Solids at 105°C Alkalinity as CaCO ₃			
Hardness Non-Filterable Residue (TSS) SM2540D Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃	☐	☐	☒
pH pH SM4500H+ B	☐	☒	☐
Settleable Solids Settleable Solids			
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C			
Simple Nutrients Ammonia as N SM4500NH ₃ D Nitrate as N ortho-Phosphate as P Nitrate + Nitrite as N	☐	☒	☐
Complex Nutrients Total phosphorus as P SM4500P E Total Kjeldahl Nitrogen	☐	☒	☐



EPA Lab Code: MI00082

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
Nitrite Nitrite as N			
Demand TOC COD BOD CBOD SM5210B	☐	☒	☐
Oil & Grease Concentrate n-Hexane Extractable Material(O&G) (Grav)			
Oil & Grease n-Hexane Extractable Material(O&G) (IR) n-Hexane Extractable Material(O&G) (Grav)			
Trace Metals Chromium Manganese Antimony Lead Molybdenum Arsenic Vanadium Zinc ✓ Beryllium Copper Aluminum Barium Boron Cadmium Iron Silver Strontium Thallium Cobalt Nickel Selenium ✗	X		
Mercury Mercury			
Low-Level Mercury Low Level Mercury			
Hexavalent Chromium Hexavalent Chromium			



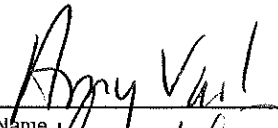
Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

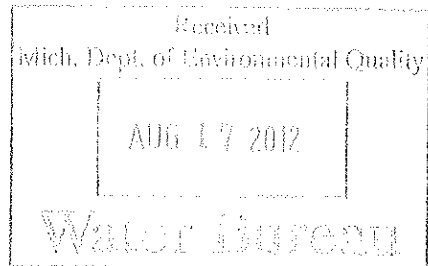
Chemistry/Microbiology Analyte Checklist DMRQA Study 32

Analyte Test / Method	Test Required	Laboratory's Graded Result	
		Acceptable	Not Acceptable (Corrective Action Required)
<u>Turbidity</u> Turbidity			
<u>Total Cyanide</u> Cyanide, total			
<u>Total Phenolics (4-AAP)</u> Phenolics, total			
<u>Total Residual Chlorine</u> Total Residual Chlorine			
<u>Low-Level Total Residual Chlorine</u> Low Level Total Residual Chlorine			
<u>WasteWatR™ Coliform MicrobE™</u> Total Coliforms (MF) Fecal Coliforms (MF) SM9222D m FC E.coli (MF) Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)	☐	☒	☐


 Print Name: Amy Vail
 Signature Title: Lab Manager

Date: 7/30/12

Use a separate checklist for EACH lab used



Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

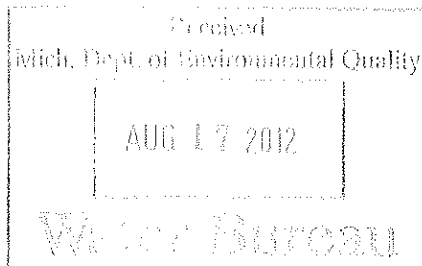
WET Organisms/Test Conditions/End Points Checklist

DMRQA Study 32

Analyte Number	Organisms / Conditions	End Points	Test Required	Laboratory's Graded Result	
				Acceptable	Not Acceptable (Corrective Action Required)
Fathead minnow (Test Code 13) 0754 48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2000.0		LC50	☐	☐	☒
Fathead minnow (Test Code 14) 0755 48Hr., Acute, Non-Renewal, 25° C, 20% DMW		LC50	☐	☐	☐
Fathead minnow (Test Code 15) 0808 7-day Short term Chronic, Daily Renewal, MHSF EPA 1000		IC25 (ON) Growth	☐	☒	☐
0810 7-day Short term Chronic, Daily Renewal, MHSF EPA 1000		NOEC (ON) Growth	☐	☒	☐
0756 7-day Short term Chronic, Daily Renewal, MHSF EPA 1000		NOEC Survival	☐	☒	☐
Fathead minnow (Test Code 16) 0812 7-day Short term Chronic, Daily Renewal, 20% DMW		IC25 (ON) Growth	☐	☐	☐
0814 7-day Short term Chronic, Daily Renewal, 20% DMW		NOEC (ON) Growth	☐	☐	☐
0759 7-day Short term Chronic, Daily Renewal, 20% DMW		NOEC Survival	☐	☐	☐
Ceriodaphnia dubia (Test Code 19) 0764 48Hr., Acute Renewal, 25° C, MHSF EPA 2002		LC50	☐	☒	☐
Ceriodaphnia dubia (Test Code 20) 0765 48Hr., Acute Renewal, 25° C, 20% DMW		LC50	☐	☐	☐
Ceriodaphnia dubia (Test Code 21) 0767 7-day Short term Chronic, Daily Renewal, MHSF EPA 1002		IC25 Reproduction	☐	☒	☐
0768 7-day Short term Chronic, Daily Renewal, MHSF EPA 1002		NOEC Reproduction	☐	☒	☐
0766 7-day Short term Chronic, Daily Renewal, MHSF EPA 1002		NOEC Survival	☐	☒	☐
Ceriodaphnia dubia (Test Code 22) 0770 7-day Short term Chronic, Daily Renewal, 20% DMW		IC25 Reproduction	☐	☐	☐
0771 7-day Short term Chronic, Daily Renewal, 20% DMW		NOEC Reproduction	☐	☐	☐
0769 7-day Short term Chronic, Daily Renewal, 20% DMW		NOEC Survival	☐	☐	☐
Daphnia magna (Test Code 32) 0788 48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2021		LC50	☐	☒	☐



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EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

Analyte Number	Organisms / Conditions	End Points	Test Required	Laboratory's Graded Result	
				Acceptable	Not Acceptable (Corrective Action Required)
Daphnia pulex (Test Code 38)					
0794	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50	☐	☐	☐
Mysid (Test Code 42)					
0798	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Inland Silverside (Test Code 45)					
0825	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0826	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0824	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐
Mysid (Test Code 43)					
0816	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0818	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0799	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐
Inland silverside (Test Code 44)					
0803	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Sheepshead minnow (Test Code 46)					
0804	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50	☐	☐	☐
Sheepshead minnow (Test Code 47)					
0820	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth	☐	☐	☐
0822	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC (ON) Growth	☐	☐	☐
0805	7-day Short term Chronic, Daily Renewal, 40 FSW	NOEC Survival	☐	☐	☐

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

SOLIDS ANALYSIS

RECEIVED

Date 3/23/12
Analyst ES

30 Minute Settling _____

SDI _____

SEP 28 2012

MLSS _____

30 Min. Settling ÷ 100 _____

Percent Removal _____

DEQ - WATER BUREAU
GRAND RAPIDS

TOTAL SUSPENDED AND TOTAL VOLATILE SUSPENDED SOLIDS (S.M. 2540D)

Sample	Wt. Fix. Liq.	Return Sl.	Raw	Primary	Secondary	Bio.	Tert.
Volume (mL)	100	50	100	50			
Cruc. & Dry Solids (g)	29.2614	29.7788	29.5769	30.4102			
Cruc. (g)	29.2660	29.7814	29.5823	30.4120			
Dry Solids (g)	.0046	.0026	.0054	.0018			
T.S.S. (mg/L)	46	50	54	36			
Cruc. & Dry Solids (g)	29.2660	29.7884	29.5823	30.4102			
Cruc. & Ash (g)							
Vol. Solids (g)							
T.V.S.S. (mg/L)							

TOTAL AND VOLATILE SOLIDS (S.M. 2540G)

Sample	Raw	V/A	Supernant	Leachate
Evap. Dish				
Dish and Wet (g)				
Dish (g)				
(A) Wet (g)				
Dish & Dry (g)				
Dish (g)				
(B) Dry (g)				
Dish & Dry (g)				
Dish & Ash (g)				
(C) Vol. Solids (g)				
TS (B+A x 100)				
TVS (C+B x 100)				